Sharp p-adic Rigidity and Source-Tunable Resonance in Stationary CCR Polynomial Transports

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October 2026

Abstract

We investigate the coefficient three degrees below the leading term of the polynomial columns associated with a stationary CCR transport recurrence. For odd orders $r \geq 5$ and primes $p > 2r$, fixing the third connected-source parameter at $\lambda_3 = 1$ forces a *sharp uniform bound* of two on its p-adic valuation, independently of all higher stationary sources. The bound is attained at $(r, p) = (7019, 21569)$. The proof reduces a putative depth-three resonance to divisibility by an explicit quadratic polynomial at one of 10,294 bounded integer parameters, followed by an exhaustive independently implemented integer-square certificate. In contrast, when λ_3 varies in $\mathbb{Z}_p$, the valuation has a complete affine-fiber classification and every positive depth, including infinite depth, occurs at a single resonant order–prime pair. We additionally describe the underlying Sheffer generating function and identify precise, restricted obstructions to direct identification with standard Nörlund and generalized Bernoulli polynomial templates. The statements concern the explicitly defined formal polynomial family, not arbitrary automaton realizations or general Stirling-number valuations.

Keywords: p-adic valuation; stationary polynomial transport; Sheffer sequence; Stirling polynomial; finite computational certificate; resonant source.

2020 Mathematics Subject Classification: Primary 11B73; Secondary 11A07, 11B68.

1 Main contributions and mathematical scope

The stationary CCR framework and its prime-adic source-refinement program were developed in the preceding article *CCR-k Arithmetic Refinement: p-Adic Resonance, Ramification, and Arithmetic Density Laws* [1]. In particular, its Section 1.1 records the normalized fixed-column transport, Section 8 analyzes the all-odd-order source resonance, and Section 12 develops the integral Stirling source obstruction. The present article is a distinct sequel: it isolates the coefficient $[X^{r-3}]F_r(X)$ and establishes its sharp good-prime valuation bound together with a complete third-source valuation law. Those two assertions are proved below rather than imported from the earlier article.

Near-diagonal Stirling arithmetic is classical, and the Nörlund and Bernoulli polynomial formalisms provide a powerful setting for its congruences; see, e.g., [2, 3, 4, 5, 8]. The present article concerns a different, explicitly specified p-dependent *stationary CCR transport*. Its fixed-column polynomials retain an ordinary Stirling shadow modulo p, but contain additional connected-source corrections with nontrivial higher p-adic behavior. Our goal is a precise theorem for one particular high coefficient of this family, not a new general theorem for ordinary Stirling numbers.

For a prime p and stationary parameters $\lambda_a \in \mathbb{Z}_p$ ($a \geq 3$), define $F_0(X) = 1$ and, for $j \geq 1$, define $F_j(X)$ by

$$\begin{aligned} T_j F_j(X) = {} & ((p+1)X - p)F_{j-1}(X) \\ & + p\big((p+1)(X-1) - p\big)F_{j-2}(X-1) \\ & + p\sum_{a=3}^{j}(-1)^a \lambda_a F_{j-a}(X-a+1), \end{aligned} \tag{1}$$

with missing negative-index terms omitted and

$$(T_j f)(X) = Xf(X+1) - (X-j)f(X).$$

The operator T_j is triangular with nonzero diagonal on $\mathbb{Q}_p[X]$ and hence invertible. Thus (1) is a self-contained definition of the polynomial family for the present article, consistent with the stationary normalized columns in [1, Sec. 1.1]. Write

$$C_{r,p}^{(\theta)} = [X^{r-3}]F_r, \qquad \lambda_3 = \theta,$$

and suppress θ when it equals 1. Higher sources λ_a, $a \geq 4$, will be arbitrary.

Theorem 1.1 (Sharp rigidity at the canonical source)**.** *For every odd integer $r \geq 5$, every prime $p > 2r$, and every choice of higher stationary sources $\lambda_a \in \mathbb{Z}_p$ ($a \geq 4$), the normalization $\lambda_3 = 1$ satisfies*

$$\boxed{0 \leq v_p(C_{r,p}) \leq 2.} \tag{2}$$

The bound is sharp: $v_p(C_{7019,21569}) = 2$.

For the second result, set

$$P(r) = 5r^2 - 45r + 64$$

and let $H(r,p)$ be the explicit integer polynomial in (5) below. The source fiber exhibits a striking contrast with Theorem 1.1.

Theorem 1.2 (Complete third-source valuation law)**.** *Fix odd $r \geq 5$ and prime $p > 2r$. Allow $\lambda_3 = \theta$ to range over $\mathbb{Z}_p$, with arbitrary fixed higher sources. Then*

(i) *If $p \nmid P(r)$, then $v_p(C_{r,p}^{(\theta)}) = 0$ for every $\theta \in \mathbb{Z}_p$.*

(ii) *If $p \mid P(r)$, the element*

$$\theta_* = 1 + \frac{H(r,p)}{17280p} \in \mathbb{Z}_p \tag{3}$$

is well defined and unique, and

$$\boxed{v_p(C_{r,p}^{(\theta)}) = 1 + v_p(\theta - \theta_*),} \tag{4}$$

with the convention $v_p(0) = +\infty$. The affine source map $\theta \mapsto C_{r,p}^{(\theta)}$ is a bijection $\mathbb{Z}_p \to p\mathbb{Z}_p$. Every positive finite valuation is attained by infinitely many ordinary integer values of θ; $\theta = \theta_*$ makes the coefficient vanish.

The main contributions are (i) a sharp, all-prime valuation obstruction for the third near-leading coefficient at the canonical source; (ii) an exact affine classification of the source fibers, showing that freeing that source permits all positive valuation depths; and (iii) a Sheffer representation providing precise comparisons with several classical polynomial templates. The finite exclusion in (i) is computer-assisted, with an explicit, reproducible certificate. The conclusions concern the stated formal stationary family, not arbitrary automaton realizations or general Stirling numbers.

The recurrence (1) is the standing definition of the polynomial family studied here. Although this formal CCR transport has a documented research antecedent in [1], the statements and proofs below are self-contained, and no claim about physical-automaton realizability is required.

2 The high-jet formula and source translation

Define

$$\begin{aligned} U_1(r) &= 35r^3 + 997r - 6816, \\ U_2(r) &= 49r^3 - 180r^2 - 925r - 384, \\ U_3(r) &= 49r^3 - 1128r^2 + 6047r - 7584, \end{aligned}$$

and let

$$H(r,p) = -(r-3)P(r) + 3pU_1(r) - 15p^2U_2(r) + 35p^3U_3(r). \tag{5}$$

Proposition 2.1 (Exact normalized high coefficient)**.** *For $r \geq 3$ and $\lambda_3 = 1$,*

$$C_{r,p} = E_{r,p}\, H(r,p), \qquad E_{r,p} = \frac{r(r-1)(r-2)(p+1)^{r-3}}{6480\, 2^r r!}. \tag{6}$$

The expression is independent of λ_a for $a \geq 4$. For $p > 2r$, the factor $E_{r,p}$ is a p-adic unit, whence

$$v_p(C_{r,p}) = v_p(H(r,p)).$$

More generally, for $\lambda_3 = \theta$ one has the exact formula

$$C^{(\theta)}_{r,p} = E_{r,p}\big[H(r,p) - 17280p(\theta-1)\big]. \tag{7}$$

Proof. Source order a enters polynomial degrees at most $r-a$, so the coefficient of X^{r-3} is independent of λ_a for $a \geq 4$. To make the coefficient induction explicit, write $F_j(X) = \sum_{k=0}^{j} f_{j,k} X^k$ and denote the right-hand side of (1) by $\mathcal{R}_j(X)$. Direct expansion of T_j gives, for $0 \leq k \leq j$,

$$(j+k) f_{j,k} + \mathbf{1}_{k\geq 1} \sum_{d=k+1}^{j} \binom{d}{k-1} f_{j,d} = [X^k]\mathcal{R}_j(X).$$

This triangular identity, together with $F_0 = 1$, determines each column from its highest coefficient downward. For $h = 0, 1, 2, 3$, normalize the four top jets by

$$J_h(j,p) = \frac{2^j j!}{(p+1)^{j-h}} [X^{j-h}] F_j.$$

Comparing powers of X in (1) gives, with $\lambda_3 = 1$,

$$\begin{aligned} J_0 &= 1, \\ J_1 &= \frac{j}{6}\big[p(7j-19) - (j-1)\big], \\ J_2 &= \frac{j(j-1)}{72}\big[(j-3)(j-2) - 2p(7j^2 + j + 42) \\ &\qquad\qquad + p^2(49j^2 - 509j + 678)\big], \\ J_3 &= \frac{j(j-1)(j-2)}{6480} H(j,p). \end{aligned}$$

Each is an identity of rational polynomials in the independent symbols (j,p); substitution into the transport recurrence, together with the initial cases, proves them by induction on j. These polynomial identities have also been checked by exact symbolic expansion. If λ_3 changes from 1 to θ, the J_0, J_1, J_2 jets are unchanged, while the difference $\delta J_3(j)$ satisfies

$$(2j-3)\delta J_3(j) - 2j\delta J_3(j-1) = -8j(j-1)(j-2)p(\theta-1).$$

Its initial value at $j = 3$ gives $\delta J_3(j) = -(8/3)j(j-1)(j-2)p(\theta-1)$, which is precisely the shift in (7). Finally, if $p > 2r$, neither $6480\, 2^r r!$ nor $r(r-1)(r-2)(p+1)$ is divisible by p; the factor $E_{r,p}$ is therefore a unit. □

3 A bounded cofactor obstruction

This section supplies the algebraic reduction that turns an *a priori* infinite collection of possible depth-three counterexamples into a finite integer certificate.

Suppose $p^3 \mid H(r,p)$ for an odd $r \geq 5$ and prime $p > 2r$. Reduction of (5) modulo p gives $p \mid P(r)$ since $0 < r-3 < p$. For $r = 5, 7, 9$, the values $P(r) = -36, -6, 64$ admit no such prime; hence $r \geq 11$. Put

$$m = \frac{P(r)}{p} \in 2\mathbb{Z}_{>0}, \qquad 0 < m < \frac{5p}{4}.$$

The parity follows directly from oddness of r, and the inequality from $r < p/2$.

Introduce

$$\begin{aligned} K(m) &= 13m^2 - 279612m + 1487901312, \\ V(m) &= -m^3 + 17616m^2 - 82615752m + 9313947648, \\ R(m) &= 76867815144960 - 8768830680m, \\ G(m,p) &= 338K(m) - pR(m). \end{aligned}$$

Lemma 3.1 (Two necessary cofactor digits)**.** *Under the depth-three hypothesis, $p \mid K(m)$ and $p^2 \mid G(m,p)$.*

Proof. Divide the original polynomial $H(X,p)$ by $P(X) - mp$. An exact integral polynomial remainder calculation gives

$$H(X,p) \equiv p(A(m,p)X + B(m,p)) \pmod{P(X) - mp},$$

where $A, B \in \mathbb{Z}[m,p]$ are printed in Appendix A. Define

$$N(m,p) = 5B^2 + 45AB + (64 - mp)A^2.$$

The Vieta identity for the roots X and $9 - X$ of $P(X) - mp$ states

$$5(AX + B)(A(9-X) + B) - N(m,p) = -A^2(P(X) - mp).$$

At $X = r$, the congruence becomes $H(r,p) = p(Ar + B)$. Hence $p^3 \mid H$ implies $p^2 \mid (Ar + B)$, and therefore $p^2 \mid N$, without needing any assumption on the conjugate root. The exactly computed expansion

$$N(m,p) = -2K(m) + pV(m) + p^2W(m,p), \qquad W \in \mathbb{Z}[m,p],$$

implies $p \mid K(m)$ and $p^2 \mid -2K(m) + pV(m)$. Polynomial division gives

$$169V(m) = (-13m - 50604)K(m) + R(m).$$

Multiplying by 169 and using $p \mid K$ now yields $p^2 \mid 338K(m) - pR(m) = G(m,p)$. □

Lemma 3.2 (Bounded integer and square obstruction)**.** *Under the same hypothesis there is an integer u such that*

$$-7667 \leq u \leq 2626, \tag{8}$$

and an admissible prime divisor $p \geq 23$, $p \notin \{29, 149\}$, of

$$Q(u) = 169u^2 + 4455261180u - 81299205874753920, \tag{9}$$

for which

$$\Delta(u,p) = (108993600 + 1040u)p^2 + 145398240p + 503620 \tag{10}$$

is a square in $\mathbb{Z}$.

Proof. We use the exact identity

$$(13m - 5220r - 116316)(13m + 5220r - 163296) = 13K(m) - 5449680P(r). \tag{11}$$

Since $p \mid P(r)$ and $p \mid K(m)$, it divides at least one factor. Let $\sigma = +1$ select the first or $\sigma = -1$ the second factor, and let u be that factor divided by p. Writing $x = 10r - 45$ gives

$$13m = up + \sigma 522x + 139806, \qquad x^2 - 745 = 20mp. \tag{12}$$

Using $0 < m/p < 5/4$, $0 < r/p < 1/2$ and $p \geq 23$ yields $-7667 \leq u \leq 16$ when $\sigma = +1$ and $-7099 \leq u \leq 2626$ when $\sigma = -1$. Thus (8) is universal.

Set

$$T_\sigma = 13(x^2 - 745) - 20p(up + \sigma 522x + 139806),$$
$$A_u = 352872u + 4651292671920, \qquad D_0 = 246462892177680.$$

A direct polynomial identity is

$$13G\left(\frac{up + \sigma 522x + 139806}{13}, p\right) = 7084584T_\sigma + p\big[p(338u^2 + 8910522360u) + \sigma A_u x + D_0\big]. \tag{13}$$

At an actual pair $T_\sigma = 0$, and Lemma 3.1 gives $p^2 \mid G$. Thus $p \mid \sigma A_u x + D_0$. But $x^2 \equiv 745 \pmod{p}$ by (12), so

$$p \mid D_0^2 - 745A_u^2 = -548913568320\, Q(u). \tag{14}$$

The fixed factor is $2^6 \cdot 3^4 \cdot 5 \cdot 13^2 \cdot 29^2 \cdot 149$. Every possible prime $p \geq 23$ dividing it is 29 or 149. Both are excluded separately: at $p = 29$ the only eligible odd base root is $r = 11$ and $p^2 \nmid H(11, 29)$; at $p = 149$ no eligible odd base root satisfies $r < p/2$. Thus $p \mid Q(u)$.

Finally, $T_\sigma = 0$ is the integer quadratic equation

$$13x^2 - \sigma 10440px - 20up^2 - 2796120p - 9685 = 0.$$

Its discriminant is precisely (10), which must be a square for integral x. □

4 Exhaustive finite exclusion and sharpness

Lemma 4.1 (Exhaustive arithmetic certificate)**.** *No integer $u \in [-7667, 2626]$ and prime $p \geq 23$, $p \notin \{29, 149\}$, satisfy both $p \mid Q(u)$ and the condition that $\Delta(u, p)$ be a perfect square.*

Computer-assisted proof. The convex quadratic $Q(u)$ is strictly negative on the stated interval (its two endpoint values are negative). For every u in this interval, the nonzero integer $|Q(u)|$ has

$$0 < |Q(u)| \leq 81{,}333{,}354{,}427{,}912{,}739 < 2^{57}.$$

Factor all 10,294 such integers, check the primality and product of their prime factors, and test (10) by exact integer square roots for every admissible prime divisor. Independent exact-arithmetic checks, using complete integer factorization, primality verification and arbitrary-precision integer square roots, agree on the following counts:

Finite-certificate quantity	Exact count
Integer parameters u	10,294
Distinct prime-factor occurrences	37,539
Eligible (u, p) records	24,248
Square discriminants among eligible records	0

The finite calculation is exhaustive because every putative counterexample must produce a prime divisor of one of the explicitly specified nonzero integers $Q(u)$; there is no empirical bound on p. The canonical serialization and SHA-256 fingerprint of the complete eligible-pair enumeration are specified in Appendix B. □

Proof of Theorem 1.1. If $v_p(C_{r,p}) \geq 3$, Proposition 2.1 gives $p^3 \mid H(r,p)$. Lemmas 3.1 and 3.2 produce (u,p) satisfying the two necessary conditions contradicted by Lemma 4.1. Hence $v_p(C_{r,p}) \leq 2$. Nonnegativity follows from (6), since $E_{r,p}$ is a p-adic unit and H is integral. At $r = 7019$ and $p = 21569$, direct evaluation of (5) satisfies $p^2 \mid H$ and $p^3 \nmid H$, and $p > 2r$. Thus the upper bound is attained. □

5 Exact source fibers and resonance engineering

Proof of Theorem 1.2. Proposition 2.1 yields

$$C_{r,p}^{(\theta)} = E_{r,p}\big[H(r,p) - 17280p(\theta - 1)\big],$$

with $E_{r,p} \in \mathbb{Z}_p^\times$. As $p > 2r \geq 10$, the integer 17280 is also a p-adic unit. Furthermore,

$$H(r,p) \equiv -(r-3)P(r) \pmod{p},$$

and $r - 3$ is a unit. If $p \nmid P(r)$, the coefficient is a unit regardless of θ. If $p \mid P(r)$, then $p \mid H$ and θ_* of (3) belongs to $\mathbb{Z}_p$. Substitution gives

$$C_{r,p}^{(\theta)} = -17280pE_{r,p}(\theta - \theta_*). \tag{15}$$

Multiplication by $-17280E_{r,p} \in \mathbb{Z}_p^\times$ proves the bijection and valuation formula. For any $k \geq 1$, choose integer $\theta \equiv \theta_* + p^{k-1} \pmod{p^k}$; then $v_p(C_{r,p}^{(\theta)}) = k$. Infinitely many such ordinary integers exist. Choosing the p-adic value $\theta = \theta_*$ makes (15) zero. □

Corollary 5.1 (Geometric depth distribution)**.** *For a fixed order–prime pair with $p \mid P(r)$, if θ is sampled according to normalized Haar measure on $\mathbb{Z}_p$, then*

$$\Pr\big(v_p(C_{r,p}^{(\theta)}) = k\big) = \frac{p-1}{p^k}, \qquad k = 1, 2, 3, \ldots.$$

The infinite-depth fiber is a singleton of Haar measure zero.

Proof. By (4), the event of depth k is $\theta - \theta_* \in p^{k-1}\mathbb{Z}_p \setminus p^k\mathbb{Z}_p$, whose Haar measure is $p^{-(k-1)} - p^{-k}$. □

Remark 5.2 (A fixed arithmetic witness)**.** *For $(r,p) = (7019, 21569)$ the canonical choice $\theta = 1$ has depth two, but the* same pair *admits every prescribed larger depth by altering θ. For example, $\theta = 281712710$ produces depth three and $\theta = 2554814402361$ produces depth four; both are checked from the exact integer polynomial H. The freedom in θ does not contradict Theorem 1.1: the canonical frozen source is its essential hypothesis.*

6 Sheffer structure and classical comparison

The fixed-column generating function admits an exact formal description which permits a precise, albeit limited, comparison with classical Bernoulli and Nörlund templates.

Proposition 6.1 (Stationary Sheffer factorization)**.** *Let* $\Phi(X,t)=\sum_{j\ge 0}F_j(X)t^j$*. There are unique formal power series* $A(t)$ *and* $B(t)$ *with* $A(0)=1$ *and* $B(0)=0$ *such that*

$$\Phi(X,t)=A(t)e^{XB(t)}.$$

They satisfy

$$e^B-1+tB'=(p+1)t+p(p+1)t^2e^{-B}, \tag{16}$$

$$t(\log A)'=-pt-p(2p+1)t^2e^{-B}+p\sum_{a\ge 3}(-1)^a\lambda_a t^a e^{(1-a)B}. \tag{17}$$

In particular, B *is independent of the connected-source parameters.*

Proof. Multiply (1) by t^j, sum over $j\ge 1$, and use the initial column $F_0=1$. One obtains

$$\begin{aligned}X(\Phi(X+1,t)-\Phi(X,t))+t\partial_t\Phi(X,t)&=((p+1)X-p)t\Phi(X,t)\\&\quad+p((p+1)(X-1)-p)t^2\Phi(X-1,t)\\&\quad+p\sum_{a\ge 3}(-1)^a\lambda_a t^a\Phi(X-a+1,t).\end{aligned}$$

Insert $\Phi=Ae^{XB}$, divide by Φ, and equate coefficients of the independent variable X; this gives (16)–(17). Both equations determine their coefficients uniquely by formal recursion; invertibility of each T_j ensures they reproduce the column recurrence. □

Writing $s=B(t)$, which is an invertible formal coordinate since $[t]B=(p+1)/2$, the source-normalized model $\lambda_3=1$ has

$$\log A(t(s))=c_1s+c_2s^2+c_3s^3+\cdots,\quad c_2=-\frac{p(5p+7)}{3(p+1)^2},\quad c_3=\frac{p(35p^2+56p-19)}{9(p+1)^3}. \tag{18}$$

Both c_2 and c_3 are nonzero for positive p. A pure binomial-type family $\exp((aX+b)g(t))$ has vanishing s^2 coefficient in this coordinate, even after an invertible formal change of t and an invertible affine change of X. Likewise the standard generalized Bernoulli family $(s/(e^s-1))^\alpha e^{Xs}$, with fixed order α, has zero cubic coefficient in $\log(s/(e^s-1))$. Consequently (18) rules out these *specific direct forms* for the CCR columns, under the indicated transformations.

These distinctions concern the stated transformations only; they are not a classification of all possible umbral equivalences or implications of earlier arithmetic theorems. In particular, Adelberg's finite-difference and Nörlund works [3, 4] are acknowledged as important related theory. We do not claim that Sheffer structure, finite-difference techniques, or Nörlund congruences originated here. Instead, the arithmetic theorem of this paper concerns the explicit p-dependent lift defined by (1) and the unusually restrictive canonical-source depth bound.

7 Computational reproducibility and scope

The computer-assisted part of the proof is only Lemma 4.1. Its finite input is defined by (9) over all integers $-7667\le u\le 2626$. Each nonzero value $|Q(u)|$ is factored exactly; every factor is checked for primality and the factorization product is reconstructed. For each eligible prime divisor, the discriminant is tested using an exact integer square root. Independent exact-integer calculations reproduce the counts in the lemma. Neither floating-point tolerances, random-prime sampling nor an unproved cutoff for counterexample primes enters the argument.

The theorem concerns formal stationary polynomial columns with specified source restrictions. It neither classifies all higher-source laws nor asserts physical automaton realizability or settles cross-order Stirling polynomial common-root problems. These distinctions are essential when comparing the present result to classical Stirling and generalized Bernoulli literature [2, 5, 6, 7, 8].

8 Related literature, originality and mathematical contribution

Classical Stirling-polynomial arithmetic and p-adic congruences have a substantial literature, including the finite-difference and Nörlund frameworks in [2, 3, 4, 5, 8]. The present article does not claim new general valuations for ordinary Stirling numbers, nor does it introduce Sheffer sequences or classical finite-difference methods. Its object is the explicitly defined, p-dependent stationary CCR transport in (1). Relative to the preceding CCR-k arithmetic-refinement study [1], the contribution isolated and proved here is an exact bound on the third near-leading coefficient at the canonical third source, and the full change in its valuation law when that source is allowed to vary. The Sheffer calculation supplies a structural comparison rather than a broad equivalence classification.

Contribution and novelty statement. The specific contributions asserted in this paper are the uniform, sharp inequality $0 \le v_p([X^{r-3}]F_r) \le 2$ for odd $r \ge 5$, $p > 2r$, and $\lambda_3 = 1$, independent of the higher stationary sources; the witness $(r, p) = (7019, 21569)$; and the exact affine source-fiber valuation law in (4). A finite arithmetic exclusion, with an explicitly specified exhaustive verification procedure and a reproducibility fingerprint, supports the uniform upper bound. No broader optimal valuation theorem for all CCR coefficients or for classical Stirling polynomials is claimed.

A Integral remainder formulas and exact identities

In Lemma 3.1, division of $H(X, p)$ by $P(X) - mp$ yields the linear remainder $p(AX + B)$, where

$$\begin{aligned} A(m,p) &= 343mp^3 - 147mp^2 + 21mp - m - 26712p^2 - 11952p + 10152, \\ B(m,p) &= -3(1603mp^3 + 261mp^2 - 63mp - m - 14112p^2 - 18624p + 10848). \end{aligned}$$

The two further identities

$$(13m - 5220r - 116316)(13m + 5220r - 163296) = 13K(m) - 5449680P(r)$$

and

$$D_0^2 - 745A_u^2 = -548913568320Q(u)$$

are integral polynomial identities; direct expansion verifies both, and the identities have additionally been checked by exact symbolic algebra.

B Reconstruction of the finite calculation

The exact finite verification can be reconstructed solely from the integer formulas in the proof. For each integer $u = -7667, -7666, \ldots, 2626$, compute $|Q(u)|$ from (9), factor it into positive prime factors, verify primality and that the product of the prime powers equals $|Q(u)|$, and retain the distinct prime divisors $p \ge 23$ other than 29 and 149. For each retained pair (u, p), calculate $\Delta(u, p)$ from (10) and check whether the nonnegative integer is a square by comparing

it with the square of its integer square root. The independently reproduced totals agree with all four exact counts in Lemma 4.1. For an exact reproducibility fingerprint, order the eligible pairs first by increasing u and, for each fixed u, by increasing p. Serialize them as UTF-8 text with an initial line `u,p`, followed by one `u,p` line per pair, with no spaces, a comma delimiter and an LF newline after every line. The SHA-256 digest of precisely this canonical serialization is (concatenate the two lines):

```
09ffb1e53a700529e73f1d9be145ad8b
794a1b0856522166d64cd1d9abc37c4c
```

The fingerprint provides an additional equality check on independently reconstructed enumerations; the exhaustiveness argument remains the exact finite reduction in Lemma 4.1.

Statements and Declarations

Author Contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, writing–original draft, and writing–review and editing.
Funding. No funding was received for conducting this study.
Competing Interests. The author has no relevant financial or non-financial interests to disclose.
Data Availability. No empirical observations were used. The finite arithmetic input, complete enumeration rules, exact summary counts and canonical SHA-256 reproducibility fingerprint are specified in the article.
Ethics Approval and Consent. Not applicable.
Consent for Publication. Not applicable.

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